DE LA SALLE UNIVERSITY-DASMARINAS

COLLEGE OF INFORMATION AND COMPUTER STUDIES

INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
Second Semester, SY 2024-2025

COURSE CODE	**:**	S-ITPC328
COURSE TITLE	**:**	System Integration and Architecture
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	Two Units
LEARNING MODALITY	**:**	Traditional Blended Learning
PRE-REQUISITES	**:**	S-ITPC316 & S-ITPC316LA
CO-REQUISITES	**:**	S-ITPC328LA
PROFESSOR	**:**	Evangeline G. Pamintuan
CONSULTATION HOURS	**:**	Monday 10:00 – 12:00 COS 100C Tuesday: 2:00 – 4:00 COS 100-C Friday 1:00 – 4:00 ONLINE
CONTACT INFORMATON	**:**	egpamintuan@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course provides an in-depth exploration of system integration and architecture. Students will learn about the concepts and best practices for designing and integrating complex systems, including topics such as system architecture, design patterns, microservices, API design, event-driven architecture, and more. Through hands-on exercises and real-world case studies, students will have the opportunity to apply their knowledge and develop their skills in this area.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. explain the fundamental concepts and principles of a system integration and architecture;
CLO2. choose common system integration patterns used to solve specific integration challenges;
CLO3. analyze the challenges and complexities associated with data integration and synchronization in heterogenous environments;
CLO4. implement the concepts of performance tuning and optimization in system integration to achieve efficient data processing and communication;
CLO5. simulate the different types od integration testing; and
CLO6. demonstrate 21st century skills in all learning activities.

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LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Explain the fundamental concepts and principles of system integration and architecture.	TLO1. Comprehend the definition and importance of system integration and architecture TLO2. Describe the various integration technologies and protocols used in different contexts.
CLO2. Choose common system integration patterns used to solve specific integration challenges	TLO3. Explain the characteristics and benefits of different system integration pattern TLO4. Understand the commonly used system integration patterns.
CLO3. Analyze the challenges and complexities associated with data integration and synchronization in Heterogenous environments.	TLO5. Assess the various data integration approaches TLO6. Examine the common challenges and complexities in system integration

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0	Week 1 Jan 20-25	• Gospel Reading, Reflection, and Prayer • Discussion of Course Syllabus **Module 0: Course Introduction**	Schoolbook	✓		2
Module 1 CLO1 TLO1 TLO2	Week 2 Jan 27-Feb 1	• Gospel Reading, Reflection, and Prayer Intro to System Integration and Architecture	Schoolbook	✓		2
	Week 3 Feb 3 - 8	• Gospel Reading, Reflection, and Prayer Enabling Assessment 1: Assessment on the key concepts of System Integration and Architecture	Schoolbook		✓	2

Module 2 CLO2 TLO3 TLO4	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer System Integration Patterns	Schoolbook	✓		2
	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer Enabling Assessment 2: Assessment on characteristic and benefits of integration patterns	Schoolbook		✓	2
Module 3 CLO3 TLO5 TLO6	Week 6 Feb 24-Mar 1	• Gospel Reading, Reflection, and Prayer Data Integration and Synchronization	Schoolbook		✓	2
	Week 7 Mar 3-8	• Gospel Reading, Reflection, and Prayer Enabling Assessment 3: Assessment of Data Integration and Synchronization concepts	Schoolbook		✓	2

	Week 8 Mar 10-15	• Gospel Reading, Reflection, and Prayer Consultation / Completion	Schoolbook			2
Midterm Exam	Week 9 Mar 17-22	• Gospel Reading, Reflection, and Prayer • Midterm Summative Assessment on Modules 1-3		✓		2
TOTAL				10	8	18

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Implement the concepts of performance tuning and optimization in system integration to achieve efficient data processing and communication.	TLO7. Explain the concepts of horizontal scaling and vertical scaling and their applicability in different integration scenarios TLO8. Apply the concept of performance tuning and optimization in system integration
CLO5. Simulate the different types of integration testing.	TLO9. Identify security risk and vulnerabilities associated with system integration TLO10. Simulate system integration, testing and validation in ensuring the proper functioning and interoperability of integrated systems
CLO6.Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self- Regulation, Real-world Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 4 CLO4 TLO7 TLO8	Week 10 Mar 24-29	• Gospel Reading, Reflection, and Prayer • **Sustainability Week** • Scalability and Performance	Schoolbook Microsoft Teams		✓	2
Module 5 CLO5 TLO9 TLO10	Week 11 Mar 31-Apr 5	• Gospel Reading, Reflection, and Prayer • • Security and Governance • in System Integration Enabling Assessment 1: Assessment on horizontal and vertical scaling	Schoolbook	✓		2
	Week 12 Apr 7-12	• Gospel Reading, Reflection, and Prayer • Enabling Assessment 2: Assessment on load balancing and resource allocation,	Schoolbook	✓		2
Module 6	Week 13 Apr 14-19	***Independent Learning Days (April 14-15)*** ***Holy Week (April 16-19)*** *Security and Governance in System Integration*			✓	2
	Week 14 Apr 21-26	***Research Week*** ***Enabling Assessment 3:*** *Assessment on security challenges, authentication and authorization*			✓	2

Module 67 CLO5 TLO9 TLO10	Week 15 Apr 28-May 3	• Gospel Reading, Reflection, and Prayer • System Integration and Validation 1		✓		2
	Week 16 May 5-10	• Gospel Reading, Reflection, and Prayer Enabling Assessment 4: Assessment on different types of testing.	Schoolbook		✓	2
	Week 17 May 12-17	• Gospel Reading, Reflection, and Prayer • Consultation /Completion Week		✓		2
Final Exam	Week 18 May 19-24	• Gospel Reading, Reflection, and Prayer • Final Term Summative Assessment on Modules 4-6		✓		2
TOTAL				10	8	18

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Provider	Reference Material
	Villegas, A., et al. 2024. System integration and architecture 1. Unlimited Books, Intramuros, Manila.
AEA Lib	Leach, Ronald J. 2016. Introduction to Software Engineering. 2nd Edition. Boca Raton. CRC Press, Taylor & Francis Group
AEA Lib	Pressman, Roger S. 2015. Software engineering: a practitioner's approach. Singapore: McGraw-Hill Book Co.

Online References

Call Number/ E-provider	Reference Material
Web Resource	AMAES TV. 2021 Jan 5. System Integration and Architecture 1 - Modeling Requirements. YouTube. [accessed 2025 Jan 3]. https://www.youtube.com/watch?v=qs4Z3PayuVQ&list=PLeggoenlMQrzUqGamlReq88emA6w-XSjS.
Web Resource	AltexSoft.2021 Mar 10. System Integration: Types, Approaches, and Implementation Steps. [accessed 2025 Jan 3] https://www.altexsoft.com/blog/system-integration/.
Web Resource	Geeky Script. 2024 Jan 30. How to Install Microsoft Visual Studio 2022 on Windows 10/11 (64 bit) [2024 Update] .NET Developers. YouTube. https://www.youtube.com/watch?v=SViilF85ues.

Prepared: Information Technology Department
2nd Semester 2024-2025

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